

MedTrack Hospital Performance Intelligence System

Dataset Column Dictionary & Tableau Dashboard Implementation Guide | Milestone 2 Reference

1. Primary Processed Datasets (Milestone_1/data/processed/)

Standardized star-schema datasets designed for Tableau without Cartesian inflation.

Table 1: hospital_overview_dataset.csv (Grain: 1 Inpatient Admission | 1,000 rows x 14 cols)

Column	Data Type	Tableau Role	Sample / Range	Description & Dashboard Role
admission_id	String	Dimension (PK)	ADM-09484753	Unique admission ID; COUNTD for admissions
patient_id	String	Dimension (FK)	PAT-09484753	Unique patient identifier
name	String	Dimension	Richard Rodriguez	Patient name for case drilldown sheets
age	Integer	Measure / Slicer	0 - 119 yrs	Age brackets (<18, 18-45, 46-65, 65+)
department_name	String	Dimension	emergency, icu, surgery, gen_med	Service ward; global slicer & treemap
admission_date	Date	Dimension (Date)	2025-01-01 to 2025-12-31	Intake date for time-series & horizons
discharge_date	Date	Dimension (Date)	2025-01-02 to 2026-01-07	Discharge date timestamp
length_of_stay_days	Integer	Measure	1 - 14 days	Duration of stay; ALOS & box-whisker
patient_satisfaction_score	Integer	Measure	60 - 99 (Avg: 80.0)	Survey score (Net Care Experience)
year, month, quarter	Integer	Dimension	2025, M1-12, Q1-4	Temporal calendar slicers
day_of_week	String	Dimension	Monday - Sunday	Weekday surge patterns
readmission_flag	Integer	Measure (Binary)	0 (0 or 1)	30-day readmission surveillance flag

Table 2: department_analytics_dataset.csv (Grain: 1 Dept x 1 Week | 208 rows x 12 cols)

Column	Data Type	Tableau Role	Sample / Range	Description & Dashboard Role
week	Integer	Dimension	1 - 52	52-week sparklines & moving averages
department_name	String	Dimension	emergency, icu, surgery, gen_med	Department identifier
patients_admitted_count	Integer	Measure	0 - 10 per wk	Weekly admitted patient throughput
avg_length_of_stay_days	Float	Measure	1.00 - 14.00 days	Weekly departmental average stay
avg_satisfaction_score	Float	Measure	60.00 - 99.00	Weekly average patient score
patients_discharged_count	Integer	Measure	0 - 10 per wk	Weekly discharge throughput
available_beds_ref	Integer	Measure	16 - 39 beds	Baseline departmental bed reference
event_ref	String	Dimension	none, flu_outbreak, strike	Operational surge/anomaly flag
bed_occupancy_rate_ref_pct	Float	Measure	0.0% - 46.4%	Derived weekly bed occupancy rate

2. Flow & Resource Datasets

Table 3: patient_flow_dataset.csv (Grain: 1 Movement Event | 2,000 rows x 10 cols)

Column	Data Type	Tableau Role	Sample / Range	Description & Dashboard Role
movement_id	String	Dimension (PK)	MOV-00001 to MOV-02000	Movement event stage ID
admission_id	String	Dimension (FK)	ADM-003ce690	Relational join to admissions
movement_type	String	Dimension	Admission, Discharge	Stage in patient progression Sankey
movement_date	Date	Dimension	2025-01-01 to 2026-01-07	Exact timestamp of movement
duration_in_department_hou rs	Float	Measure	24.0 - 336.0 hrs	Hours in department (LOS x 24)

Table 4: resource_utilization_dataset.csv (Grain: 1 Dept x 1 Wk x 1 Resource | 628 rows x 8 cols)

Column	Data Type	Tableau Role	Sample / Range	Description & Dashboard Role
resource_utilization_id	String	Dimension (PK)	RES-00001 to RES-00628	Unique resource log ID
week	Integer	Dimension	1 - 52	Operational week number
resource_type	String	Dimension	doctor, nurse, bed	Resource role filter
resource_category	String	Dimension (Param)	staff, bed	Drives Bed vs Staff parameter toggle
total_units_available	Integer	Measure	1 - 39 units	Operational capacity assigned
units_in_use	Integer	Measure	1 - 39 units	Active units deployed in shift
utilization_rate_pct	Float	Measure	25.0% - 100.0%	Resource strain percentage (Workforce Stress)

3. Milestone 2 Scorecard & Novel KPIs (Milestone_2/data/)

KPI ID & Name	Category	Value	Benchmark	Novel / Differentiated Advantage
KPI-01: Total Admissions	Volume	1,000	1,000	Standard baseline throughput
KPI-02: Avg Length of Stay	Operational	7.41 days	< 7.00 days	CMS/National benchmark bullet graph
KPI-03: Bed Occupancy Rate	Capacity	20.47%	80.0% - 85.0%	Licensed capacity baseline
KPI-04: Bed Utilization Rate	Capacity	92.47%	85.0%	Emergency at 100% capacity
KPI-07: Patient Turnaway Rate	Volume	56.64%	< 20.0%	Novel KPI 1: Unserved community demand (7,642 refused)
KPI-08: Demand Fulfillment Rate	Volume	43.36%	> 80.0%	Novel KPI 2: Inverse fulfillment ratio
KPI-09: Net Care Experience	Clinical	80.0 / 100	> 85.0	Novel KPI 3: Patient satisfaction distribution
KPI-10: Staff Morale Score	Workforce	72.57 / 100	> 75.0	Novel KPI 4: Caregiver sentiment index
KPI-11: Clinical Workforce Stress	Workforce	88.98%	< 80.0%	Novel KPI 5: Doctor & nurse saturation percentage

4. Tableau Dashboard Visual Mapping Summary

Dashboard	Visual Name	Chart Type	Primary Dataset Fields
Dashboard 1: Hospital Overview	Demand Horizon Ribbon	Dual-Axis Area Ribbon	week, patients_request, patients_admitted, patients_refused
Dashboard 1: Hospital Overview	Department Intake Volume	Hierarchical Treemap	department_name, admission_id (COUNTD), week
Dashboard 1: Hospital Overview	Bed Occupancy Waffle	100-Mark Matrix & Radial	department_name, bed_utilization_pct
Dashboard 1: Hospital Overview	ALOS Target Benchmark	Clinical Bullet Graph	department_name, length_of_stay_days (AVG), Target: 7.0d
Dashboard 2: Patient Flow	Clinical Progression Flow	Patient Acuity Sankey	movement_type, department_name, admission_id (COUNTD)
Dashboard 2: Patient Flow	Length of Stay Variance	Box-and-Whisker Plot	department_name, length_of_stay_days
Dashboard 3: Bed & Resources	Strategic Efficiency Matrix	4-Quadrant Scatter Plot	X: bed_utilization, Y: demand_fulfillment, Size: stress_index

Dashboard 3: Bed & Resources	Capacity Thresholds	Safety Bands Bullet Chart	resource_category, utilization_rate_pct, Bands: 80%/85%/95%
Dashboard 3: Bed & Resources	Ward Strain Temporal Grid	Heatmap Calendar Matrix	day_of_week, resource_type, utilization_rate_pct